

Web Technology Programs



HOME LIBS TABLE IMAGES

DEPARTMENT OF COMPUTER SCIENCE & IT

Submitted by :-
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MCA^{4th} Sem Subjects

- **Web Technologies**
 - HTML, CSS, JAV
 - JavaScript
 - Servlets and JSP
 - ASP Controls
 - State Management in ASP
- **Machine Learning using Python**
 - Introduction to Machine Learning
 - Types of Machine Learning
 - Statistical Algorithm and Deep Learning
 - Introduction to Python
 - Introduction to Machine Learning using Python and Deep Learning
- **Theory of Computation and Compiler Design**
 - Regular Languages and Expressions
 - Context Free Grammar
 - Turing Machines
 - Compiler Structure & Frontend
 - Error Detection and Recovery, Code Generation & Optimization

MCA 3rd Semester Time Table

Day	9:30 AM – 10:30 AM	10:30 AM – 12:30 PM	12:30 PM – 1:30 PM	1:30 PM – 2:30 PM	2:30 PM – 3:30 PM	3:30 PM – 4:30 PM
Monday	Web Technologies	Software Engineering	Theory of Computation and compiler design	Break	Machine learning using Python	Practical
Tuesday	Web Technologies	Software Engineering	Theory of Computation and compiler design	Break	Big data analytics using H	Practical
Wednesday	Big data analytics using H	Theory of Computation and compiler design	Software Engineering	Break	Machine learning using Python	Practical
Thursday	Web Technologies	Big data analytics using H	Theory of Computation and compiler design	Break	Machine learning using Python	Practical
Friday	Web Technologies	Software Engineering	Machine learning using Python	Break	Big data analytics using H	Practical

Teaching/Non-Teaching Record Table

Faculty Name	Designation	Area of Interest
Dr. Shubhrodran Singh Jarnail	Professor	Java & Web Technologies
Dr. Jasbir Singh	Assistant Professor	Theory of Computation & CA
Dr. Anil Kumar	Programmer	C++ & Python
Dr. Sangeeta Khatun	Technical Staff	Workshop
Mr. Ajay Khatun	Non-Teaching Staff	Department

Images



1. Design front static page for department:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
```

```
<title>Department of Computer Science
& IT - University of Jammu</title>
```

```
<style>
```

```
@import
```

```
url('https://fonts.googleapis.com/css2?family=J
aro:opsz@6..72&family=Poppins:ital,wght@0,1
00;0,200;0,300;0,400;0,500;0,600;0,700;0,800
;0,900;1,100;1,200;1,300;1,400;1,500;1,600;1,
700;1,800;1,900&display=swap');
```

```
*{
font-family: 'Poppins',Arial, sans-serif;
margin: 0;
padding: 0 1px;
}
```

```
header {
background-color: #f4f4f4;
padding: 7px 0;
text-align: center;
border-radius: 7px;
margin-bottom: 2px;
}
```

```
header img{
width: 300px;
border-radius: 12px;
}
```

```
nav{
background-color: #f4f4f4;
padding: 16px;
text-align: center;
margin-bottom: 7px;
}
```

```
nav a {
margin: 0 20px;
text-decoration: none;
```

```
color: #6d6dd2;
font-weight: bold;
}
```

```
nav a:hover{
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}
nav a:active{
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}
```

```
h1{
text-align: center;
color: #060f12;
font-family: "Jaro",Arial, sans-serif;
}
```

```
h3{
color: #6885a7;
}
li{
margin: 2px 17px;
}
```

```
.student-details{
background-color: #f4f4f4;
text-align: center;
line-height: 1.5;
}
.syllabus-part, .semester-div{
background-color: #f4f4f4;
margin: 20px 0;
padding: 7px 30px;
max-width: auto;
}
.syllabus-part h3{
text-align: center;
}
```

```
table{
width: 100%;
border-collapse: collapse;
margin: 20px 0;
}
```

```
table,
```

```

th,
td {
border: 1px solid #ddd;
}

th,
td {
padding: 7px;
text-align: left;
}

th {
background-color: #6d6dd2;
color: white;
}
.dept-imgs img{
width: 240px;
display: block;
margin: 10px auto;
}
</style>
</head>

<body>
<header>

<nav>
<a href="#">HOME</a>
<a href="#">LISTS</a>
<a href="#">TABLES</a>
<a href="#">IMAGES</a>
</nav>
<main>
<h1><span>DEPARTMENT OF COMPUTER
SCIENCE &IT </span></h1>
<div class="student-details">
<h3>Submitted by 1 </h3>
<p>Himanshu Parihar | 30-MCA-23</p>
<p>Agrim Sharma | 16-MCA-23</p>
<p>Kapil Sadotra | 13-MCA-23</p>
<p>Atul Kohli | 11-MCA-23</p>
<p>Prabhjeet Choudhary | 20-MCA-23</p>
</div>
<div class="syllabus-part">
<h3>MCA 3<sup>RD</sup> Sem Subjects</h3>

```

```

<ul>
<li><strong>Web Technologies</strong>
<ul>
<li>HTML/CSS/XML</li>
<li>JavaScript</li>
<li>Servlets and JSP</li>
<li>ASP Controls</li>
<li>State Management in ASP</li>
</ul>
</li>
<li><strong>Machine Learning Using
Python</strong>
<ul>
<li>Introduction to Machine
Learning</li>
<li>Types of Machine Learning</li>
<li>Genetic Algorithm and Deep
Learning</li>
<li>Introduction to Python</li>
<li>Introduction to Machine Learning
using Python and Deep Learning</li>
</ul>
</li>
<li><strong>Theory of Computation and
Compiler Design</strong>
<ul>
<li>Regular Languages and
Expressions</li>
<li>Context Free Grammar</li>
<li>Turing Machines</li>
<li>Compiler Structure & Frontend</li>
<li>Errors Detection and Recovery,
Code Generation & Optimization</li>
</ul>
</li>
</ul>
</div>
<div class="semester-div">
<h3>MCA 3rd Semester Time Table</h3>
<table>
<thead>
<tr>
<th>Day</th>
<th>10:00 AM - 11:00 AM</th>
<th>11:00 AM - 12:00 PM</th>
<th>12:00 PM - 1:00 PM</th>

```

```

<th>1:00 PM - 1:30 PM</th>
<th>1:30 PM - 2:30 PM</th>
<th>2:30 PM - 4:30 PM</th>
</tr>
</thead>
<tbody>
<tr>
<td>Monday</td>
<td>Web Technologies</td>
<td>Software Engineering</td>
<td>Theory of Computation and compiler
design</td>
<td>Break</td>
<td>Machine learning using Python</td>
<td>Practical</td>
</tr>
<tr>
<td>Tuesday</td>
<td>Web Technologies</td>
<td>Software Engineering</td>
<td>Theory of Computation and
compiler design</td>
<td>Break</td>
<td>Big data analytics using R</td>
<td>Practical</td>
</tr>
<tr>
<td>Wednesday</td>
<td>Big data analytics using R</td>
<td>Theory of Computation and
compiler design</td>
<td>Software Engineering</td>
<td>Break</td>
<td>Machine learning using Python</td>
<td>Practical</td>
</tr>
<tr>
<td>Thursday</td>
<td>Web Technologies</td>
<td>Big data analytics using R</td>
<td>Theory of Computation and
compiler design</td>
<td>Break</td>
<td>Machine learning using Python</td>
<td>Practical</td>
</tr>
<tr>
<td>Friday</td>

```

```

<td>Web Technologies</td>
<td>Software Engineering</td>
<td>Machine learning using Python</td>
<td>Break</td>
<td>Big data analytics using R</td>
<td>Practical</td>
</tr>
</tbody>
</table>
<h3>Teaching/Non-Teaching Record
Table</h3>
<table>
<thead>
<tr>
<th>Faculty Name</th>
<th>Designation</th>
<th>Area of Interest</th>
</tr>
</thead>
<tbody>
<tr>
<td>Dr. Shubhnandan Singh Jamwal</td>
<td>Professor</td>
<td>Java & Web Technologies</td>
</tr>
<tr>
<td>Sh. Jasbir Singh</td>
<td>Assistant Professor</td>
<td>Theory of Computation & ADA</td>
</tr>
<tr>
<td>Sh. Anil Tickoo</td>
<td>Programmer</td>
<td>C++ & Python</td>
</tr>
<tr>
<td>Sh. Sanjay Manhas</td>
<td>Technical Staff</td>
<td>Workshop</td>
</tr>
<tr>
<td>Mr. Ajay Malgotra</td>
<td>Non-Teaching Staff</td>
<td>Department</td>
</tr>
</tbody>
</table>
</div>

```

```

<div class="dept-imgs">
<h3>Images</h3>

</div>
</main>
</body>
</html>

```

2. WAP to find the greatest of three numbers (user defined input) using ternary operators:

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Find Greatest Number</title>
<style>
body{
font-family: Arial, sans-serif;
text-align: center;
margin-top: 50px;
}
input, button{
margin: 10px;
padding: 10px;
font-size: 16px;
}
#output{
margin-top: 20px;
font-size: 18px;
color: green;
}
</style>
</head>
<body>
<h1>Find the Greatest Number</h1>
<label for="num1">Enter first
number:</label>
<input type="number" id="num1"
placeholder="First Number"><br>
<label for="num2">Enter second
number:</label>

```

```

<input type="number" id="num2"
placeholder="Second Number"><br>
<label for="num3">Enter third
number:</label>
<input type="number" id="num3"
placeholder="Third Number"><br>
<button onclick="findGreatest()">Find
Greatest</button>
<div id="output"></div>
<script>
function greatest(a, b, c){
let value = a > b && a > c ? a
: b > a && b > c ? b
: c;
return value;
}
function findGreatest(){
let num1 =
document.getElementById('num1').value;
let num2 =
document.getElementById('num2').value;
let num3 =
document.getElementById('num3').value;
let result = greatest(num1, num2, num3);
document.getElementById('output').innerT
ext = `${result} is the greatest`;
}
</script>
</body>
</html>

```

Find the Greatest Number

Enter first number:

Enter second number:

Enter third number:

777 is the greatest

- Write HTML code to create login form and additional tags form as shown in the figure using the div and required tags:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Practical 2</title>
<style>
body{
font-family: Arial, sans-serif;
margin: 0;
padding: 0;
background-color: #f9f9f9;
```

```
}

.container {
width: 90%;
max-width: 800px;
margin: 20px auto;
padding: 20px;
background-color: #ffffff;
border: 1px solid #ddd;
border-radius: 8px;
box-shadow: 0 2px 4px rgba(0, 0, 0,
0.1);
}
```

```
h1,
h2 {
text-align: center;
color: #004080;
}
```

```
form {
margin-top: 20px;
}
```

```
.form-group {
margin-bottom: 15px;
}
```

```
.form-group label {
display: block;
margin-bottom: 5px;
font-weight: bold;
}
```

```
.form-group input,
.form-group select,
.form-group textarea {
width: 100%;
padding: 8px;
border: 1px solid #ccc;
border-radius: 4px;
}
```

```
.form-group input[type="radio"] {
width: auto;
}
```

```
.form-actions {
```

```
text-align: center;
margin-top: 20px;
}
```

```
.form-actions button {
padding: 10px 20px;
margin: 0 10px;
border: none;
border-radius: 4px;
cursor: pointer;
font-size: 16px;
}
```

```
.form-actions button.submit {
background-color: #004080;
color: white;
}
```

```
.form-actions button.reset {
background-color: #f44336;
color: white;
}
</style>
</head>
```

```
<body>
<div class="container">
<h1>HTML FORMS</h1>
<h2>LOGIN FORM</h2>
<form>
<div class="form-group">
<label for="username">User
Name:</label>
<input type="text" id="username"
name="username" placeholder="Enter your
username" required>
</div>
<div class="form-group">
<label for="lastname">Last
Name:</label>
<input type="text" id="lastname"
name="lastname" placeholder="Enter your last
name" required>
</div>
<div class="form-actions">
<button type="submit"
class="submit">SUBMIT</button>
```

```
<button type="reset"
class="reset">RESET</button>
</div>
</form>
<h2>ADDITIONAL TAGS - BASIC FORM</h2>
<form>
<fieldset>
<legend>PERSONAL DETAILS</legend>
<div class="form-group">
<label for="firstname">First
Name:</label>
<input type="text" id="firstname"
name="firstname" placeholder="Enter your first
name" required>
</div>
<div class="form-group">
<label>Gender:</label>
<input type="radio" id="male" name="gender"
value="Male" required>
<label for="male">Male</label>
<input type="radio" id="female" name="gender"
value="Female">
<label for="female">Female</label>
</div>
<div class="form-group">
<label for="dob">DOB
(dd/mm/yyyy):</label>
<input type="date" id="dob" name="dob"
required>
</div>
<div class="form-group">
<label for="email">Email:</label>
<input type="email" id="email" name="email"
placeholder="Enter your email" required>
</div>
<div class="form-group">
<label for="contact">Contact:</label>
<input type="tel" id="contact" name="contact"
placeholder="Enter your contact number"
required>
</div>
<div class="form-group">
<label for="address">Address:</label>
<textarea id="address" name="address"
rows="4" placeholder="Enter your
address"></textarea>
</div>
<div class="form-group">
```

```

<label>Qualification:</label>
<input type="checkbox" id="qualification10"
name="qualification" value="10th">
<label for="qualification10">10th</label>
<input type="checkbox" id="qualification12"
name="qualification" value="12th">
<label for="qualification12">12th</label>
<input type="checkbox" id="graduation"
name="qualification" value="Graduation">
<label for="graduation">Graduation</label>
</div>
<div class="form-group">
<label for="identity">Identity
Proof:</label>
<select id="identity" name="identity" required>
<option value="">Select Identity
Proof</option>
<option value="aadhar">Aadhar
Card</option>
<option value="pan">PAN Card</option>
<option value="passport">Passport</option>
</select>
</div>
<div class="form-actions">
<button type="submit"
class="submit">SUBMIT</button>
<button type="reset"
class="reset">RESET</button>
</div>
</fieldset>
</form>
</div>
</body>
</html>

```

4. Define external styling sheet:

- a. To style paragraph using id names "para1".

Styles.css file

```

#para1{
color: green;
}

```

- b. Add stylesheet (internal/ external) in Question 1.

Internal Stylesheet –

Already used with <style> tag

External Stylesheet –
Style.css

```

*{
font-family: 'Poppins',Arial, sans-serif;
margin: 0;
padding: 0 1px;
}

```

```

header {
background-color: #f4f4f4;
padding: 7px 0;
text-align: center;
border-radius: 7px;
margin-bottom: 2px;
}
header img{
width: 300px;
border-radius: 12px;
}

```

```

nav {
background-color: #f4f4f4;
padding: 16px;
text-align: center;
margin-bottom: 7px;
}

```

```

nav a {
margin: 0 20px;
text-decoration: none;
color: #6d6dd2;
font-weight: bold;
}

```

```

nav a:hover {
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}
nav a:active{
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}

```

```

h1{

```



```

text-align: center;
color: #060f12;
font-family: "Jaro",Arial, sans-serif;
}
h3{
color: #6885a7;
}
li{
margin: 2px 17px;
}
.student-details{
background-color: #f4f4f4;
text-align: center;
line-height: 1.5;
}
.syllabus-part, .semester-div{
background-color: #f4f4f4;
margin: 20px 0;
padding: 7px 30px;
max-width: auto;
}
.syllabus-part h3{
text-align: center;
}

table{
width: 100%;
border-collapse: collapse;
margin: 20px 0;
}

table,
th,
td{
border: 1px solid #ddd;
}

th,
td{
padding: 7px;
text-align: left;
}
th{
background-color: #6d6dd2;
color: white;
}
.dept-imgs img{
width: 240px;

```

```

display: block;
margin: 10px auto;
}

```

5. WAP to Implement the Methods of Math Class:

```

<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Math Class Methods</title>
<script>
function showMathMethods(){
let num = 25.6;
let result = `
Math.ceil(${num}) =
${Math.ceil(num)}<br>
Math.floor(${num}) =
${Math.floor(num)}<br>
Math.round(${num}) =
${Math.round(num)}<br>
Math.sqrt(16) = ${Math.sqrt(16)}<br>
Math.pow(2, 3) = ${Math.pow(2, 3)}<br>
Math.abs(-5) = ${Math.abs(-5)}<br>
Math.min(3, 7, 1) = ${Math.min(3, 7,
1)}<br>
Math.max(3, 7, 1) = ${Math.max(3, 7,
1)}<br>
Math.random() =
${Math.random().toFixed(2)}<br>`;
document.write(result)
}
</script>
</head>

<body>
<h1>Math Class Methods</h1>
<button onclick="showMathMethods()">Show
Results</button>
<div id="mathResults"></div>
</body>

</html>

```

```

Math.ceil(25.6) = 26
Math.floor(25.6) = 25
Math.round(25.6) = 26
Math.sqrt(16) = 4
Math.pow(2, 3) = 8
Math.abs(-5) = 5
Math.min(3, 7, 1) = 1
Math.max(3, 7, 1) = 7
Math.random() = 0.14

```

6. Paragraph Analysis with Frequency, Unique Words, and Replacement:

```

<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Paragraph Analysis</title>
<script>
function analyzeText() {
let text =
document.getElementById("inputText").value;
let words =
text.toLowerCase().match(/\b\w+\b/g);
let frequency = {};
let longestUniqueWord = "";
let highestFrequencyWord = "";
let maxFrequency = 0;
words.forEach(word => {
frequency[word] = (frequency[word] || 0) + 1;
if (word.length > longestUniqueWord.length &&
frequency[word] === 1) {
longestUniqueWord = word;
}
if (frequency[word] > maxFrequency) {
maxFrequency = frequency[word];
highestFrequencyWord = word;
}
});

```

```

let replacedText = text.replace(/\b\w+\b/gi,
"ComputerScienceDepartmentUniversityOf
Jammu");
document.getElementById("output").innerHTM
L = `<strong>Unique Words and
Frequency:</strong><br>${JSON.stringify(frequ
ency, null, 2)}<br>
<strong>Longest Unique Word:</strong>
${longestUniqueWord}<br>
<strong>Highest Frequency
Word:</strong>
${highestFrequencyWord}<br>
<strong>Replaced
Text:</strong><br>${replacedText}<br>`;
}
</script>
</head>

```

```

<body>
<h1>Paragraph Analysis</h1>
<textarea id="inputText" rows="10" cols="80"
placeholder="Enter your paragraph
here..."></textarea>
<br>
<button onclick="analyzeText()">Analyze</butto
n>
<div id="output"></div>
</body>
</html>

```

Paragraph Analysis

Hum to doobe hain unki tasveeron mein
Mile sakoon nahi unke bin neendon mein
Kya yaad mujhe bhi karti hogi woh
Kya yaad mujhe bhi karti hogi woh

Ab uska pata nahi dil mein jo rehna tha
Uski baaton ke afaane likhne baithe
Kya yaad mujhe bhi karti hogi woh
Kya yaad mujhe bhi karti hogi woh

Analyze

Unique Words and Frequency:

("hum": 8, "to": 10, "doobe": 5, "hain": 9, "unki": 5, "tasveeron": 5, "mein": 20, "mile": 5, "sakoon": 5, "nahi": 31, "unke": 5, "bin": 9, "neendon": 5, "kya": 27, "yaad": 17, "mujhe": 17, "bhi": 21, "karti": 18, "hogi": 18, "woh": 24, "ab": 6, "uska": 3, "pata": 5, "dil": 6, "jo": 7, "rehta": 3, "tha": 4, "uski": 3, "baaton": 3, "ke": 6, "afsaane": 3, "likhne": 3, "baithe": 3, "haan": 1, "maane": 4, "kar": 1, "yeh": 6, "khata": 1, "tiya": 1, "mera": 1, "sake": 1, "galti": 2, "chhupake": 1, "saare": 1, "black": 1, "ko": 5, "kiya": 5, "white": 1, "mere": 5, "bin": 1, "sne": 1, "s": 1, "dying": 1, "falaun": 1, "main": 8, "adwayen": 1, "kaali": 1, "sara": 1, "kahna": 1, "par": 4, "tu": 10, "kahani": 1, "jaise": 5, "miss": 2, "karun": 2, "baby": 1, "you": 2, "me": 1, "birthday": 1, "bola": 1, "tune": 3, "wish": 1, "dino": 1, "strong": 1, "ha": 15, "bata": 2, "ye": 2, "tuhse": 2, "kisne": 1, "kaha": 2, "khair": 1, "jhelna": 1, "padega": 1, "pain": 1, "hisse": 1, "ka": 1, "purane": 1, "apne": 1, "chats": 1, "read": 2, "type": 1, "karke": 1, "messages": 1, "delete": 1, "puchhna": 1, "chahi": 1, "ki": 4, "kaisa": 1, "d": 2, "try": 1, "kar": 1, "rahi": 1, "are": 1, "trying": 1, "reach": 1, "tere": 7, "khayaal": 1, "hoon": 2, "khyaal": 1, "rakhta": 2, "saath": 3, "hoda": 2, "malaal": 1, "paas": 1, "hua": 1, "just": 1, "lost": 1, "wapis": 1, "pyaar": 2, "karne": 1, "majaal": 1, "naa": 3, "khayaal": 1, "tera": 1, "se": 1, "mitaya": 1, "abhi": 2, "gayi": 3, "fir": 1, "lagaya": 1, "kabhi": 4, "bewafa": 2, "tuhiko": 1, "bulaya": 1, "bas": 2, "ginwaya": 1, "bass": 1, "ginni": 1, "barsaat": 1, "aur": 2, "diye": 1, "gun": 1, "maane": 1, "baat": 1, "bhole": 1, "baba": 1, "karo": 1, "in": 1, "kashton": 1, "kam": 1, "kaali": 1, "raat": 2, "ek": 1, "poori": 2, "bottle": 1, "rum": 1, "sober": 1, "daaru": 1, "badi": 1, "pee": 1, "mohabbat": 1, "khayalon": 1, "hi": 2, "jee": 1, "mare": 1, "rahe": 1, "milne": 1, "koshish": 1, "meri": 2, "royi": 1, "roti": 1, "khayi": 1, "doston": 1, "bola": 2, "baare": 1, "chhod": 1, "banda": 1, "khada": 1, "wahin": 1, "galat": 1, "ladki": 1, "sahi": 1, "yeh": 1, "dikhe": 2, "dard": 1, "mahaul": 1, "sard": 1, "aansu": 1, "aankhein": 1, "kiski": 1, "rota": 2, "mard": 1, "kaun": 1, "hamare": 1, "kamre": 1, "sota": 1, "lighten": 1, "band": 1, "iv": 1, "on": 2, "hota": 2, "old": 1, "school": 1, "mujhse": 1, "move": 1, "jaane": 2, "pehle": 1, "lade": 1, "pehli": 1, "dafa": 1, "aisi": 1, "huyi": 1, "khafa": 1, "aisa": 1, "kyon": 1, "pichhle": 1, "mahine": 1, "candle": 1, "har": 1, "show": 1, "kal": 1, "light": 1, "yun": 1, "bathe": 2, "ro": 1, "diya": 2, "pe": 1, "chala": 1, "kho": 1, "rahun": 3, "dooba": 12, "sa": 12, "soona": 12, "bhoola": 12, "lagun": 1, "more": 1, "https": 1, "lyricsmint": 1, "com": 1, "ikka": 1 }

Longest Unique Word: lyricsmint

Highest Frequency Word: nahi

Replaced Text:

Hum to doobe hain unki tasveeron mein Mile sakoon nahi unke bin neendon mein Kya yaad mujhe bhi karti hogi woh Ab uska pata nahi dil mein jo rehna tha Uski baaton ke afaane likhne baithe Kya yaad mujhe bhi karti hogi woh Hum to doobe hain unki tasveeron mein Mile sakoon nahi unke bin neendon mein Kya yaad mujhe bhi karti hogi woh Haan maine kari yeh khata liya mera side Galti chhupake saare black ko kiya white Mere bina she's dying falaun main atwayen Kaafi sara kehna par pata nahi tu kahan hai Jaise miss main karun baby you miss me kya Mere birthday mein beta tune wish nahi kiya Dino strong hai bata ye tuhe kisne kaha Khair ab to jhelna padega pain ye hisse ka Kya purane apne chats ko tu karti hogi read Kya tu type karke messages karti hogi delete Kya tu bhi puchhna chahi hai ki kaisa hai tu d Kya tu bhi try kar nahi hai are you trying to reach Tere khayaal mein nahi hoon par tu khyaal rakhta D saath mein nahi thoda malaal rakhta Jo bhi tha mere paas woh hua hai just lost Wapis pyaar karne ki karun majaal ab naa Hum to doobe hain unki tasveeron mein Mile sakoon nahi unke bin neendon mein Kya yaad mujhe bhi karti hogi woh Kya yaad mujhe bhi karti hogi woh Khayaal tera dil se mitaya nahi abhi Tu gayi fir dil yeh lagaya nahi kabhi Bewafa maine tujhko bulaya nahi abhi Pyaar kiya bas tuhe ginwaya nahi kabhi Bass ginni barsaat aur tere diye gum Dil maane nahi baat ke ab saath nahi hum Bhole baba thoda karo in kashton ko kam Mere saath kaali raat aur ek poori bottle rum Pooni raat nahi sober daaru badi pee hai Yeh jo mohabbat khayalon mein hi jee hai Hum to maare jaa rahe hain milne ko Kya tune bhi milne ki koshish kabhi ki hai Meri yaad mein tu royi roti khayi tune nahi hai Doston ko bola mere baare bola nahi hai Chhod ke tu gayi par banda khada wahin hai Galat bas hum woh to ladki hai to sahi hai Hum to doobe hain unki tasveeron mein Mile sakoon nahi unke bin neendon mein Kya yaad mujhe bhi karti hogi woh Kya yaad mujhe bhi karti hogi woh Yeah, Jo dikhe naa woh dard nahi Kya Mahaul yeh sard hai Kya Aansu naa dikhe aankhein hain kiski Jo rota hai woh mard nahi Kya Yeh bata mujhe ke kaun nahi rota Main hamare kamre mein nahi sota Lighten band tu on nahi hota Main hoon old school Mujhse move on nahi hota Kya jaane woh kaha gayi Pehle hi lade hain yeh pehli dafa nahi Par kabhi aisi huyi woh khata nahi Galti hai meri woh bewafa nahi Maine aisa kyon kiya Pichhle mahine candle har show kiya Kal flight mein yun hi bathe bathe ro diya Tere jaane pe pata chala maine kya kho diya Hum to doobe hain unki tasveeron mein Mile sakoon nahi unke bin neendon mein Kya yaad mujhe bhi karti hogi woh Kya yaad mujhe bhi karti hogi woh Ab uska pata nahi dil mein jo rehna tha Uski baaton ke afaane likhne baithe Kya yaad mujhe bhi karti hogi woh Rahun main jaise Dooba dooba dooba sa Soona soona soona sa Bhoola bhoola bhoola sa Tere bin, Rahun main jaise Dooba dooba dooba sa Soona soona soona sa Bhoola bhoola bhoola sa Tere bin Read soona sa Bhoola bhoola bhoola sa Tere bin, Lagun main jaise Dooba dooba dooba sa Soona soona soona sa Bhoola bhoola bhoola sa

more: <https://lyricsmint.com/ikka/woh>

7. Gmail Registration form with validation:

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport"

content="width=device-width, initial-scale=1.0">

<title>Gmail Registration Form</title>

<style>

body{

padding: 13px 13px;

}

input{

margin: 7px 0;

}

</style>

<script>

function validateForm() {

const fname =

document.getElementById("fname").value.trim(

);

const lname =

document.getElementById("lname").value.trim(

);

const email =

document.getElementById("email").value.trim()

;

const password =

document.getElementById("password").value.trim();

const confirmPassword =

document.getElementById("confirmPassword").value.trim();

if (!fname || !lname || !email || !password || !confirmPassword) {

alert("All fields are required.");

return false;

}

if (password !== confirmPassword) {

alert("Passwords do not match.");

return false;

}

const emailPattern = /^[a-zA-Z0-9._-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,6}\$/;

if (!emailPattern.test(email)) {

alert("Invalid email format.");

return false;

}

alert("Registration Successful!");

return true;

}

</script>

</head>

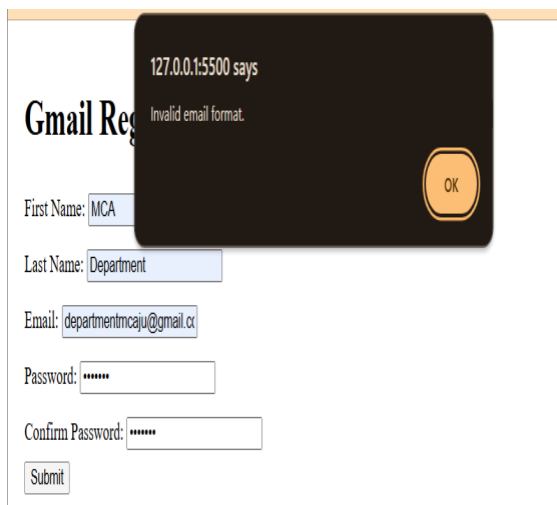
```
<body>
<h1>Gmail Registration Form</h1>
<form onsubmit="return validateForm()">
First Name: <input type="text" id="fname"><br>
Last Name: <input type="text" id="lname"><br>

Email: <input type="email" id="email"><br>

Password:      <input      type="password"
id="password"><br>

Confirm Password: <input type="password"
id="confirmPassword"><br>
<button type="submit">Submit</button>
</form>
</body>

</html>
```



The screenshot shows a web browser displaying a Gmail Registration Form. A dark brown modal dialog box is overlaid on the form, indicating an error. The dialog box contains the text "127.0.0.1:5500 says" and "Invalid email format." with an "OK" button. The form fields are as follows:

- First Name: MCA
- Last Name: Department
- Email: departmentmcaju@gmail.co
- Password: [masked with asterisks]
- Confirm Password: [masked with asterisks]
- Submit button

8. Find and Replace Operation:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport"
content="width=device-width, initial scale=1.0"
/>
<title>Find and Replace</title>
<script>
function findReplace() {
let text =
document.getElementById("inputText").value;
let words = text.match(/\b\w+\b/g) || [];
let foundWords = words.filter(word =>
/[aprs]/i.test(word));
let replacedText = text.replace(/\b[aprs]\w*/gi,
"123CSDEPARTMENT");
document.getElementById("output").innerHTM
L = `
<strong>Found Words:</strong>
${foundWords.join(", ")}<br>
<strong>Replaced Text:</strong>
${replacedText}`;
}
</script>

</head>
<body>
<h1>Find and Replace</h1>
<textarea id="inputText" rows="10" cols="80"
placeholder="Enter your text
here..."></textarea>
<br>
<button onclick="findReplace()">Find &
Replace</button>
<div id="output"></div>
</body>
</html>
```

Find and Replace

As we explained before, supervised learning is a type of machine learning where a model is trained on labeled data—meaning each input is paired with the correct output. The model learns by comparing its predictions with the actual answers provided in the training data. Over time, it adjusts itself to minimize errors and improve accuracy. The goal of supervised learning is to make accurate predictions when given new, unseen data. For example, if a model is trained to recognize handwritten digits, it will use what it learned to correctly identify new numbers it hasn't seen before.

Find & Replace

Found Words: As explained before, supervised learning is a type of machine learning where a model is trained on labeled data—meaning each input is paired with the correct output. The model learns by comparing its predictions with the actual answers provided in the training data. Over time, it adjusts itself to minimize errors and improve accuracy. The goal of supervised learning is to make accurate predictions when given new, unseen data. For example, if a model is trained to recognize handwritten digits, it will use what it learned to correctly identify new numbers it hasn't seen before.

Replaced Text: 123CSDEPARTMENT

The goal of 123CSDEPARTMENT learning is to make 123CSDEPARTMENT when given new, unseen data. For example, if 123CSDEPARTMENT model is trained to 123CSDEPARTMENT handwritten digits, it will use what it learned to correctly identify new numbers it hasn't 123CSDEPARTMENT before.

9. Age Calculations with DOB:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width,
initial-scale=1.0">
<title>Age Calculator</title>
</head>
<body>
<h1>Age Calculator</h1>
<form>
<label for="dob">Enter Your DOB:</label>
<input id="dob" type="date">

<button type="button" id="cal-
btn">Calculate</button>
<button type="reset" id="rst-
btn">Reset</button>

<div id="age-result"></div>
</form>

<script>
let dob = document.getElementById('dob');
let calculate = document.getElementById('cal-
btn');
let ageResult = document.getElementById('age-
result');
let rstbtn = document.getElementById('rst-btn');

calculate.addEventListener('click', function() {
if (dob.value) {
let birthDate = new Date(dob.value);
let currentDate = new Date();
let age = currentDate.getFullYear() -
birthDate.getFullYear();
let monthDiff = currentDate.getMonth() -
birthDate.getMonth();

if (monthDiff < 0 || (monthDiff === 0 &&
currentDate.getDate() < birthDate.getDate())) {
age--;
}
}
```

```
ageResult.textContent = `Your age is: ${age}
years old.`;
}else{
ageResult.textContent = 'Please select a valid
date of birth.';
}
});
</script>
```

```
</body>
</html>
```

Age Calculator

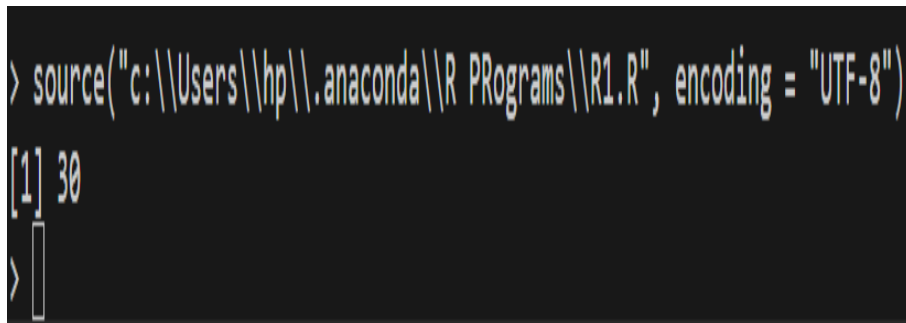
Enter Your DOB: 

Your age is: 23 years old.

R Programs

1. WAP to display any value in vector from location:

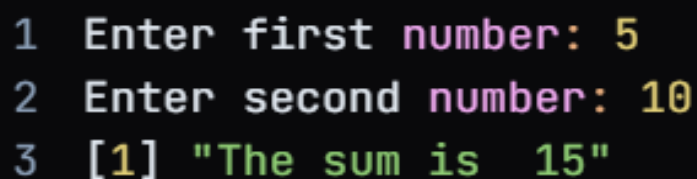
```
my_vector <- c(10, 20, 30, 40, 50)
index <- 3
value_at_index <- my_vector[index]
print(value_at_index)
```

A terminal window with a dark background. The first line shows the command to source a file: > source("c:\\Users\\hp\\.anaconda\\R Programs\\R1.R", encoding = "UTF-8"). The second line shows the output: [1] 30. The third line shows the prompt > followed by a cursor.

```
> source("c:\\Users\\hp\\.anaconda\\R Programs\\R1.R", encoding = "UTF-8")
[1] 30
> 
```

2. WAP to add two numbers using functions.

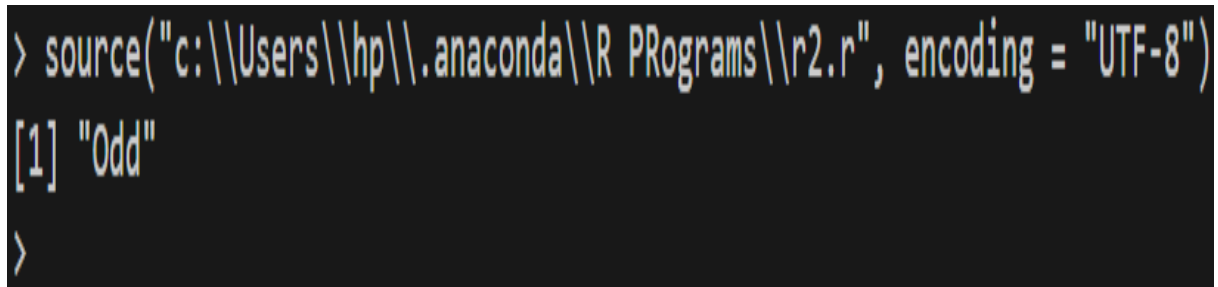
```
twoNumAdder <- function(num1, num2){
  return(num1 + num2)
}
number1 <- as.integer(readline(prompt = "Enter first number: "))
number2 <- as.integer(readline(prompt = "Enter second number: "))
paste("The sum is ",twoNumAdder(number1, number2))
```

A terminal window with a dark background. It shows three lines of input and output. Line 1: 1 Enter first number: 5. Line 2: 2 Enter second number: 10. Line 3: 3 [1] "The sum is 15".

```
1 Enter first number: 5
2 Enter second number: 10
3 [1] "The sum is 15"
```

3. WAP to check if a number is even or odd by using function.

```
is_even <- function(num) {  
  if (num %% 2 == 0) {  
    return("Even")  
  } else {  
    return("Odd")  
  }  
}  
  
number <- 25  
result <- is_even(number)  
print(result)
```



```
> source("c:\\Users\\hp\\.anaconda\\R Programs\\r2.r", encoding = "UTF-8")  
[1] "Odd"  
>
```

4. Create a list, add an element, and remove the last element.

```
my_list <- list(1, 2, 3, 4, 5)  
my_list <- append(my_list, 6)  
my_list <- my_list[-length(my_list)]  
print(my_list)
```



```
> source("c:\\Users\\hp\\.anaconda\\R Programs\\R3.r", encoding = "UTF-8")
[[1]]
[1] 1

[[2]]
[1] 2

[[3]]
[1] 3

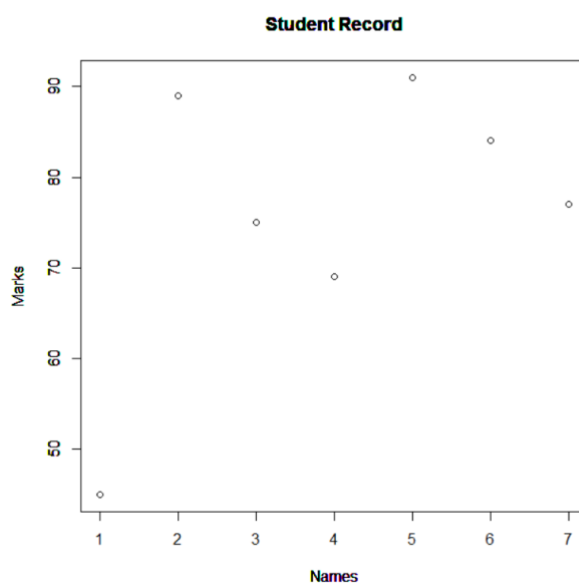
[[4]]
[1] 4

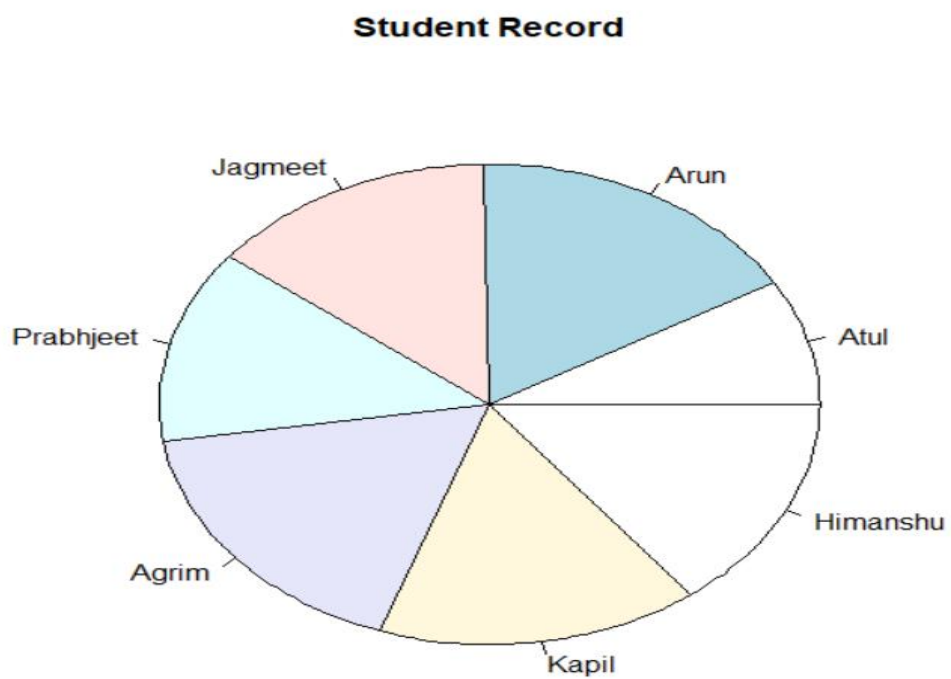
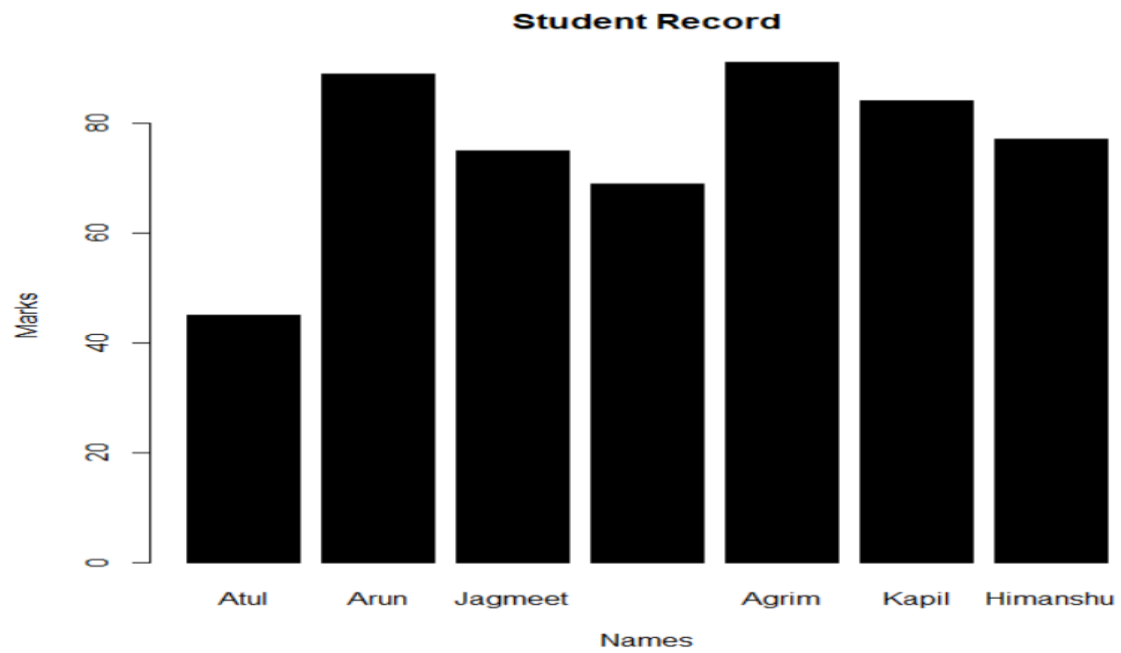
[[5]]
[1] 5
```

5. To create a data frame. Display plot, bar-plot and pie chart.

```
friends <- data.frame(
  id = c(1:7),
  names = c("Atul","Arun","Jagmeet","Prabhjeet","Agrim","Kapil","Himanshu"),
  marks = c(45,89,75,69,91,84,77)
)
print(friends)
plot(friends$marks, xlab="Names",ylab="Marks", main="Student Record")
barplot(friends$marks, xlab="Names",ylab="Marks", main="Student Record", names.arg=friends$names,col="black")
pie(friends$marks, labels=friends$names,main="Student Record")
```

```
> print(friends)
  id  names marks
1  1   Atul   45
2  2   Arun   89
3  3 Jagmeet   75
4  4 Prabhjeet 69
5  5   Agrim   91
6  6   Kapil   84
7  7 Himanshu   77
> |
```





6. WAP to create data frame for 10 students containing rollno, name & marks of 3 subjects - sub1, sub2 & sub3.

```
df <- data.frame(
```

```
Roll_No = 1:10,
```

```

Name = c("Agrim", "Bobby", "Prabjeet", "Mansi", "Himanshu", "Atul", "Etika",
"Mona", "Raj", "Kapil"),

Sub1 = c(85, 92, 78, 95, 88, 79, 90, 82, 93, 87),

Sub2 = c(90, 88, 92, 85, 91, 86, 93, 89, 87, 94),

Sub3 = c(88, 91, 87, 90, 89, 85, 92, 88, 90, 86)

)

print(df)

```

```

> source("c:\\Users\\hp\\.anaconda\\R Programs\\r4.r", encoding = "UTF-8")
  Roll_No  Name Sub1 Sub2 Sub3
1      1  Agrim   85   90   88
2      2  Bobby   92   88   91
3      3 Prabjeet  78   92   87
4      4  Mansi   95   85   90
5      5 Himanshu  88   91   89
6      6   Atul   79   86   85
7      7  Etika   90   93   92
8      8  Mona   82   89   88
9      9   Raj   93   87   90
10     10 Kapil   87   94   86
>

```

7. Generate dataset with 10 rows and 4 columns(ID, Name, Age, Score). Include atleast 5 NA values distributed across different columns. Print the original dataset.

```

df <- data.frame(

  ID = 1:10,

  Name = c("Agrim", "Himanshu", "Prabjeet", "Mansi", "Manisha", "Atul", "Etika",
"Mona", "Raj", "Kapil"),

  Age = c(20, 22, 21, NA, NA, 24, 25, 26, NA, 23),

  Score = c(85, 92, 78, 78, NA, 99, NA, NA, 82, 86)

)

print(df)

```

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	78
4	4	Mansi	NA	78
5	5	Manisha	NA	NA
6	6	Atul	24	99
7	7	Etika	25	NA
8	8	Mona	26	NA
9	9	Raj	NA	82
10	10	Kapil	23	86

Write Code to:

- Remove all rows that contain NA values from the dataset and print the cleaned dataset.

```
df_cleaned <- na.omit(df)
print(df_cleaned)
```

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	78
6	6	Atul	24	99
10	10	Kapil	23	86

- Replace NA values in numeric columns (Age & Score) with median of the respective column and print the dataset

```
median_age <- median(df$Age, na.rm = TRUE)
```

```
median_score <- median(df$Score, na.rm = TRUE)
```

```
df$Age[is.na(df$Age)] <- median_age
```

```
df$Score[is.na(df$Score)] <- median_score
```

```
print(df)
```

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	78
4	4	Mansi	23	78
5	5	Manisha	23	85
6	6	Atul	24	99
7	7	Etika	25	85
8	8	Mona	26	85
9	9	Raj	23	82
10	10	Kapil	23	86

8. Perform slice and filter operation on dataset of program

```
# Slice Operation
```

```
sliced_data <- df[3:6, c("Name", "Score")]
```

```
print(sliced_data)
```

	Name	Score
3	Prabjeet	78
4	Mansi	78
5	Manisha	NA
6	Atul	99

```
# Filter Operation
```

```
filtered_data <- df[df$Score >= 85, ]
```

```
print(filtered_data)
```

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	88
6	6	Atul	24	99
10	10	Kapil	23	86

9. From the given data set:
- Calculate the covariance
 - Calculate the correlation coefficient
 - Interpret the results

```
study_hours <- c(2,4,6,8,10)
exam_scores <- c(50,55,60,70,85)
covariance <- cov(study_hours, exam_scores)
```

```
correlation_coefficient <- cor(study_hours, exam_scores)
covariance
correlation_coefficient
```

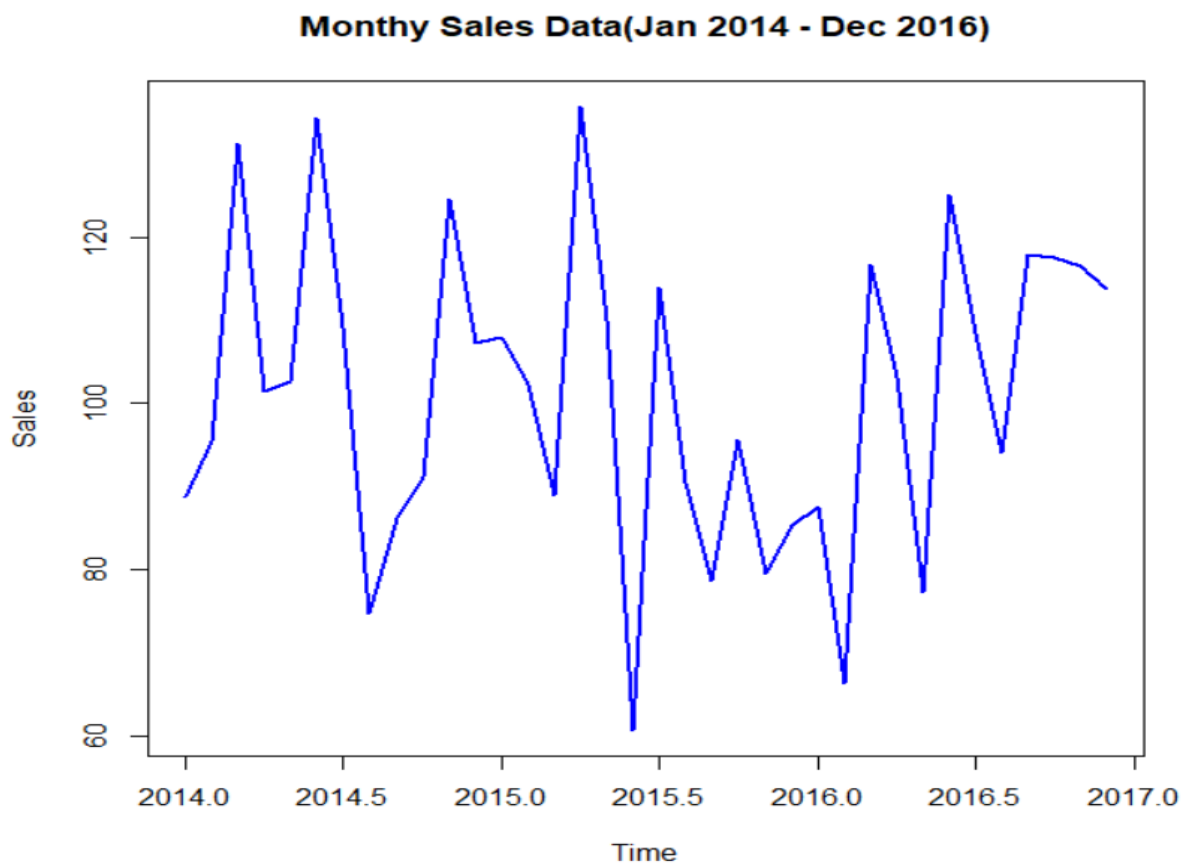
```
>
> covariance
[1] 42.5
> correlation_coefficient
[1] 0.9686649
> |
```

10. Create a time series object.

```
set.seed(123)
sales_data <- rnorm(36, mean=100, sd=20)
```

```
sales_ts <- ts(sales_data, start=c(2014,1),frequency=12)
```

```
plot(sales_ts, main="Monthy Sales Data(Jan 2014 - Dec  
2016)",xlab="Time", ylab="Sales", col="blue", lwd=2)
```



Python Programs

1. WAP to swap two numbers

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

print()

print("Before swap")

print("Number1 = ", num1, " & Number2 = ", num2)

temp = num1
num1 = num2
num2 = temp

print()

print("After swap")

print("Number1 = ", num1, " & Number2 = ", num2)
```

```
● python-programs-sem3 ➤ python -u "c:\Code\Python\python-programs-sem3\one.py"

Enter first number: 7

Enter second number: 0

Before swap
Number1 = 7 & Number2 = 0

After swap
Number1 = 0 & Number2 = 7
```


2. WAP to check whether the given number is even or odd using conditional statement.

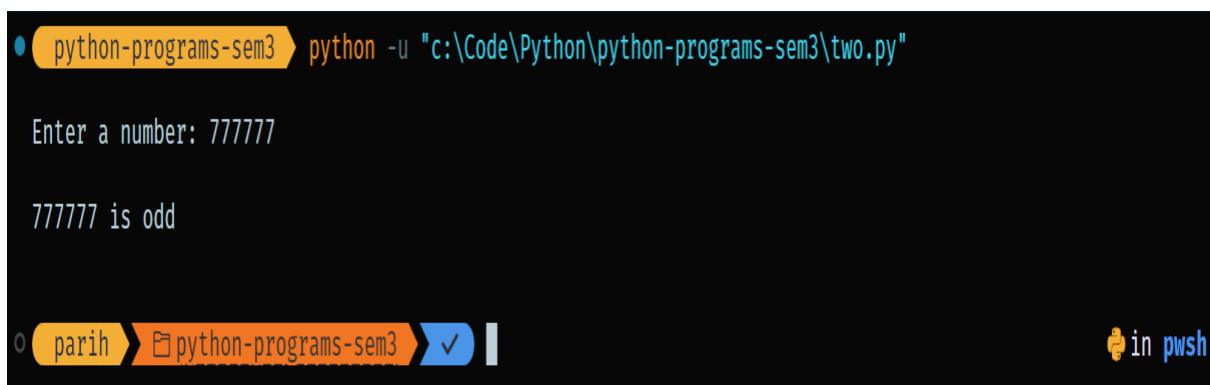
```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0 :
```

```
    print(num, "is even")
```

```
else:
```

```
    print(num, "is odd")
```



A terminal window with a dark background. The prompt is 'python-programs-sem3'. The command entered is 'python -u "c:\Code\Python\python-programs-sem3\two.py"'. The output shows 'Enter a number: 777777' followed by '777777 is odd'. The terminal path is 'parih > python-programs-sem3' and it shows a green checkmark icon. The bottom right corner has 'in pwsh'.

3. WAP to check whether the given year is leap year or not using conditional statement.

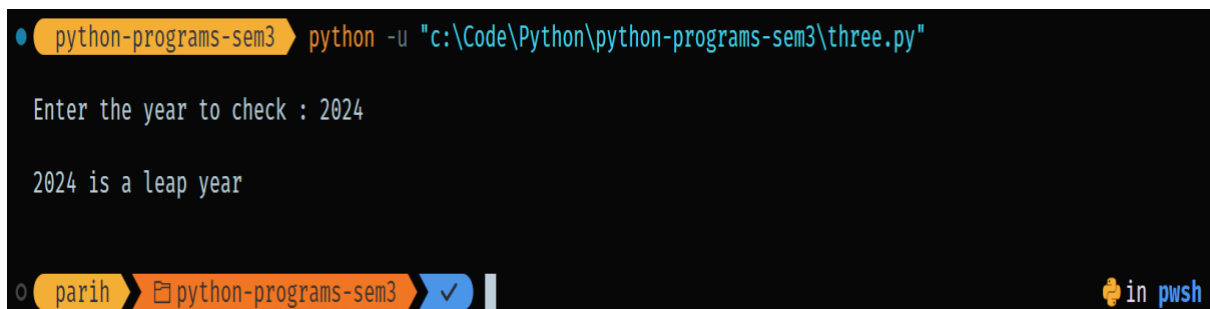
```
year = int(input("Enter the year to check : "))
```

```
if (year % 400 == 0) and (year % 4 == 0) or (year % 100 != 0):
```

```
    print(year, "is a leap year")
```

```
else:
```

```
    print(year, "is not a leap year")
```



A terminal window with a dark background. The prompt is 'python-programs-sem3'. The command entered is 'python -u "c:\Code\Python\python-programs-sem3\three.py"'. The output shows 'Enter the year to check : 2024' followed by '2024 is a leap year'. The terminal path is 'parih > python-programs-sem3' and it shows a green checkmark icon. The bottom right corner has 'in pwsh'.

4. WAP to find max. number from list of no's using loops

```
numbers_list = [33,22,45,67,89,9,1,23,100]
```

```
maxNum = numbers_list[0]
```

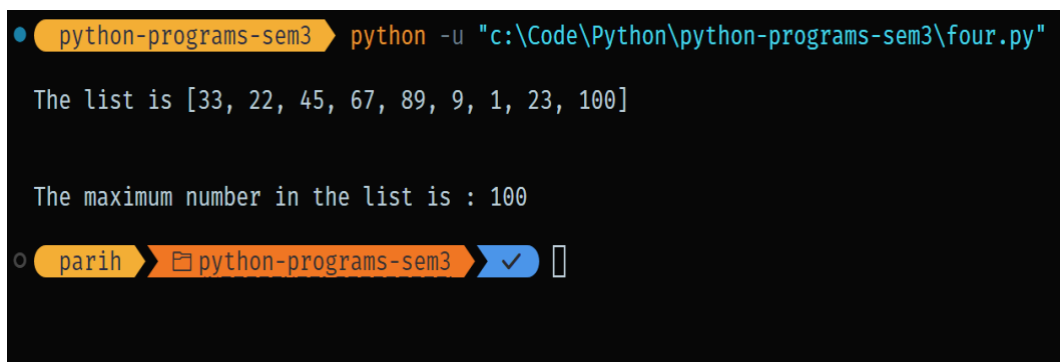
```
print("The list is", numbers_list,)
```

```
for num in numbers_list:
```

```
    if num > maxNum:
```

```
        maxNum = num
```

```
print("The maximum number in the list is :", maxNum)
```



```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\four.py"

The list is [33, 22, 45, 67, 89, 9, 1, 23, 100]

The maximum number in the list is : 100

parih > python-programs-sem3 > [checkmark]
```

5. WAP to calculate sum of even no's from 1 to 100 including 1 & 100 using range function.

```
start = 1
```

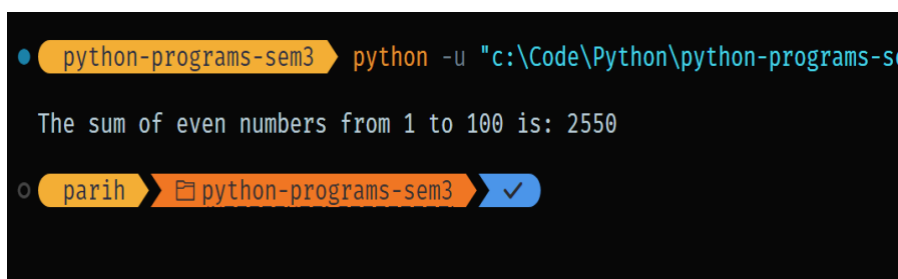
```
end = 100
```

```
sum = 0
```

```
for num in range(start, end + 1, 2):
```

```
    sum += num
```

```
print("The sum of even numbers from 1 to 100 is:", sum)
```



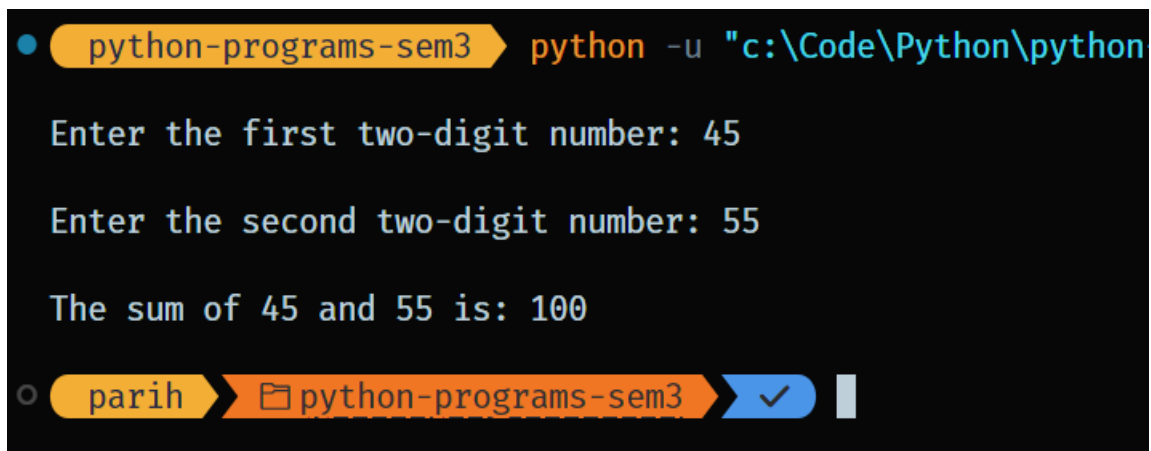
```
python-programs-sem3 python -u "c:\Code\Python\python-programs-s...

The sum of even numbers from 1 to 100 is: 2550

parih > python-programs-sem3 > [checkmark]
```

6. WAP to add sum of two-digit numbers.

```
def checkTwoDigit(num):  
    if num < 10 or num > 99:  
        print("Please enter a valid two-digit number.")  
        return False  
    return True  
  
first_number = int(input("Enter the first two-digit number: "))  
second_number = int(input("Enter the second two-digit number: "))  
  
if (checkTwoDigit(first_number) and checkTwoDigit(second_number)):  
    print(f"The sum of {first_number} and {second_number} is: {first_number +  
second_number}")  
else: print("Both numbers should be two-digit numbers.")
```



```
python-programs-sem3 python -u "c:\Code\Python\python  
Enter the first two-digit number: 45  
Enter the second two-digit number: 55  
The sum of 45 and 55 is: 100  
parih python-programs-sem3 ✓
```

7. WAP to convert variables from one datatype to another.

```
def type_conversion_demo():  
    int_var = 10  
    float_var = 20.5  
    str_var = "30"  
    converted_float = float(int_var)
```

```

print(f"Converted int to float: {converted_float} (Type: {type(converted_float)})")
converted_int = int(float_var)
print(f"Converted float to int: {converted_int} (Type: {type(converted_int)})")
converted_str_to_int = int(str_var)
print(f"Converted string to int: {converted_str_to_int}
(Type:{type(converted_str_to_int)})")
converted_int_to_str = str(int_var)
print(f"Converted int to string: '{converted_int_to_str}' (Type:
{type(converted_int_to_str)})")
str_float = "45.67"
converted_str_to_float = float(str_float)
print(f"Converted string to float: {converted_str_to_float} (Type:
{type(converted_str_to_float)})")
list_var = [1, 2, 3]
converted_list_to_str = str(list_var)
print(f"Converted list to string: '{converted_list_to_str}' (Type:
{type(converted_list_to_str)})")
converted_str_to_list = eval(converted_list_to_str)
print(f"Converted string back to list: {converted_str_to_list} (Type:
{type(converted_str_to_list)})")
type_conversion_demo()

```

python-programs-sem3 ➤ python -u "c:\Code\Python\python-program

```

Converted int to float: 10.0 (Type: <class 'float'>)
Converted float to int: 20 (Type: <class 'int'>)
Converted string to int: 30 (Type: <class 'int'>)
Converted int to string: '10' (Type: <class 'str'>)
Converted string to float: 45.67 (Type: <class 'float'>)
Converted list to string: '[1, 2, 3]' (Type: <class 'str'>)
Converted string back to list: [1, 2, 3] (Type: <class 'list'>)

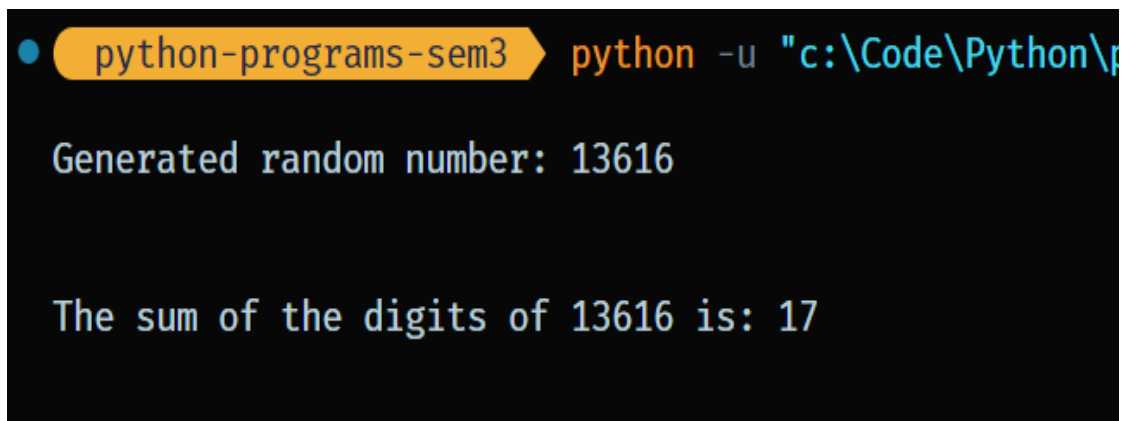
```

8. WAP to generate random numbers and then calculate its sum of digits.

```
import random

def sum_of_digits(number):
    total = 0
    while number > 0:
        digit = number % 10
        total += digit
        number //= 10
    return total

random_number = random.randint(1, 99999)
print(f"Generated random number: {random_number}")
digit_sum = sum_of_digits(random_number)
print(f"The sum of the digits of {random_number} is: {digit_sum}")
```



```
python-programs-sem3 python -u "c:\Code\Python\p...
Generated random number: 13616
The sum of the digits of 13616 is: 17
```

9. WAP to generate first 50 prime number series. Also check whether given any two no's are coprime or not.

```
import math

def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(math.sqrt(n)) + 1):
```

```
    if n % i == 0:  
        return False  
    return True
```

```
def generate_primes(count):  
    primes = []  
    num = 2  
    while len(primes) < count:  
        if is_prime(num):  
            primes.append(num)  
        num += 1  
    return primes
```

```
def gcd(a, b):  
    while b:  
        a, b = b, a % b  
    return a
```

```
def are_coprime(a, b):  
    return gcd(a, b) == 1
```

```
first_50_primes = generate_primes(50)  
print("First 50 prime numbers:")  
print(first_50_primes)  
num1 = int(input("Enter the first number: "))  
num2 = int(input("Enter the second number: "))  
  
if are_coprime(num1, num2):
```

```
print(f"{num1} and {num2} are coprime.")
```

else:

```
print(f"{num1} and {num2} are not coprime.")
```

```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\nine.py"
First 50 prime numbers:

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229]

Enter the first number: 19
Enter the second number: 29
19 and 29 are coprime.
```

10.WAP to display Fibonacci series.

```
num = int(input("Enter a number upto which you want to display the fibonacci series: "))
```

```
n1, n2 = 0, 1
```

```
print("Fibonacci Series:", n1, n2, end=" ")
```

```
for i in range(2, num):
```

```
    n3 = n1 + n2
```

```
    n1 = n2
```

```
    n2 = n3
```

```
    print(n3, end=" ")
```

```
print()
```

```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\ten.py"

Enter a number upto which you want to display the fibonacci series: 12

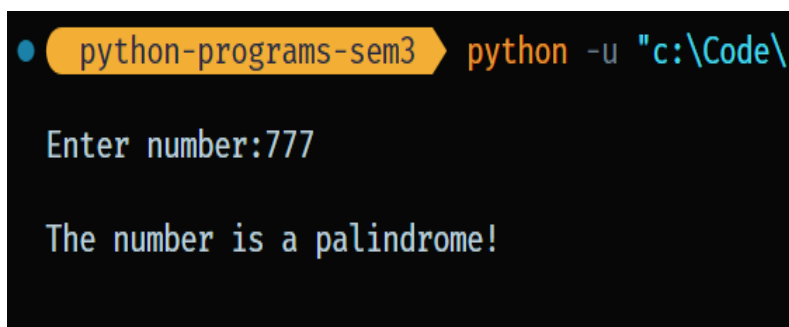
Fibonacci Series: 0 1
1 2 3 5 8 13 21 34 55 89
```

11.WAP to check whether a number is palindrome or not.

```

n=int(input("Enter number:"))
temp=n
rev=0
while(n>0):
    dig=n%10
    rev=rev*10+dig
    n=n//10
if(temp==rev):
    print("The number is a palindrome!")
else:
    print("The number isn't a palindrome!")

```



```

python-programs-sem3 python -u "c:\Code\l...
Enter number:777
The number is a palindrome!

```

12.WAP to perform the following operations on string:

- a. find index of particular character.

```

text = input("Enter the string: ")
char_to_find = input("Enter the character to find: ")

if char_to_find in text:
    index = text.index(char_to_find)
    print(f"The index of '{char_to_find}' is: {index}")
else:
    print(f"'{char_to_find}' not found in the string.")

```



```
python-programs-sem3 > python -u "c:\Code\Python\
Enter the string: Machine Learning using Python
Enter the character to find: P
The index of 'P' is: 23
```

b. perform string uppercase and lowercase.

```
print("String in lowercase:", text.lower())
print("String in uppercase:", text.upper())
```

```
Enter the string: Python
String in lowercase: python
String in uppercase: PYTHON
```

c. count the occurrence of particular character in string.

```
text = input("Enter the string: ")
char = input("Enter the character to find and count: ")
if char in text:
    index = text.index(char)
    print(f"The index of '{char}' is: {index}")
else:
    print(f"'{char}' not found in the string.")
count = text.count(char)
print(f"Count of '{char}' in '{text}': {count}")
```

```
parih > python-programs-sem3 > python -u "c:\o
on\python-programs-sem3\twelve.py"

Enter the string: Machine Learning using Python
Enter the character to find and count: n
The index of 'n' is: 5
Count of 'n' in 'Machine Learning using Python': 5
```

13.WAP to perform operations on sets.

```
s1 = {1, 2, 3, 4, 5}
s2 = {4, 5, 6, 7, 8}
print("Union of sets: ", s1.union(s2))
print("Intersection of sets: ", s1.intersection(s2))
print("Symmetric Difference: ", s1.symmetric_difference(s2))
print("Are sets disjoint: ", s1.isdisjoint(s2))
```

```
ms-sem3\thirteen.py"
Union of sets:  {1, 2, 3, 4, 5, 6, 7, 8}
Intersection of sets:  {4, 5}
Symmetric Difference:  {1, 2, 3, 6, 7, 8}
Are sets disjoint:  False
```

14.WAP to perform Operations on Tuples.

```
x = ('a', 'b', 'c', 'b')
y = ('d', 'e', 'f')
print("Count of b: ", x.count('b'))
print("Position of c: ", x.index('c'))
```

```
z = x+y
print("Joining two tuples:", z)
l = list(x)
print(type(l))
```

```
Count of b: 2
Position of c: 2

Joining two tuples: ('a', 'b', 'c', 'b', 'd', 'e', 'f')
<class 'list'>
```

15.WAP to perform operations on List.

```
list1 = [1, 2, 3, 4, 5, 6]
list2 = [7, 8, 9, 10, 11]
print("Joining two lists: ", list1+list2)
list2.append(12)
print("After appending: ", list2)
list1.reverse()
print("After reversing: ", list1)
x = [3, 15, 4, 1, 2]
x.sort()
print("After Sorting: ", x)
print("Maximum element of list: ", max(list1))
```

```
Joining two lists:  [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

After appending:  [7, 8, 9, 10, 11, 12]

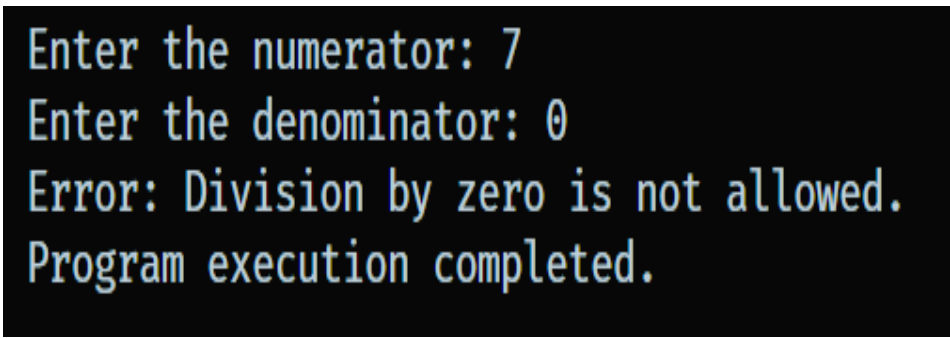
After reversing:  [6, 5, 4, 3, 2, 1]

After Sorting:  [1, 2, 3, 4, 15]

Maximum element of list:  6
```

16.WAP for Exception Handling.

```
try:
    num1 = int(input("Enter the numerator: "))
    num2 = int(input("Enter the denominator: "))
    result = num1 / num2
    print(f"The result is: {result}")
except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")
except ValueError:
    print("Error: Invalid input. Please enter numeric values.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
finally:
    print("Program execution completed.")
```



```
Enter the numerator: 7
Enter the denominator: 0
Error: Division by zero is not allowed.
Program execution completed.
```

17. WAP to demonstrate Naïve Bayes.

```
!pip install pandas
!pip install numpy
!pip install scikit-learn
import pandas as pd
import numpy as np
import sklearn
import os

cwd = os.getcwd()
print(cwd)
dataset=pd.read_csv('heart.csv' , engine='python')
with open('heart.csv') as fp:
    data = fp.read().split('\n')

print(dataset)
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	\
0	52	1	0	125	212	0	1	168	0	1.0	
1	53	1	0	140	203	1	0	155	1	3.1	
2	70	1	0	145	174	0	1	125	1	2.6	
3	61	1	0	148	203	0	1	161	0	0.0	
4	62	0	0	138	294	1	1	106	0	1.9	
...	...	...	..	...	...	...	...	...	...	...	
1020	59	1	1	140	221	0	1	164	1	0.0	
1021	60	1	0	125	258	0	0	141	1	2.8	
1022	47	1	0	110	275	0	0	118	1	1.0	
1023	50	0	0	110	254	0	0	159	0	0.0	
1024	54	1	0	120	188	0	1	113	0	1.4	

	slope	ca	thal	target
0	2	2	3	0
1	0	0	3	0
2	0	0	3	0
3	2	1	3	0
4	1	3	2	0
...	...	..	...	...
1020	2	0	2	1
1021	1	1	3	0
1022	1	1	2	0
1023	2	0	2	1
1024	1	1	3	0

[1025 rows x 14 columns]

```
X = dataset.drop('target',axis=1)
```

```
Y = dataset['target']
```

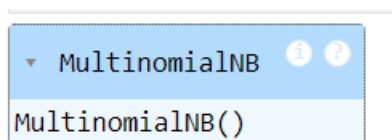
```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test,Y_train, Y_test = train_test_split(X,Y, test_size = 0.30)
```

```
from sklearn.naive_bayes import MultinomialNB
```

```
classifier=MultinomialNB()
```

```
classifer.fit(X_train,Y_train)
```



```
print(classifier.predict(X[:]))
```

[1 0 0 ... 0 1 0]

```
Y_pred=classifier.predict(X_test)
from sklearn.metrics import accuracy_score
```

```
print (accuracy_score(Y_test,Y_pred))
```

```
0.7435064935064936
```

```
from sklearn.metrics import confusion_matrix
```

```
cm=confusion_matrix(Y_test,Y_pred)
print(cm)
```

```
[[105  43]
 [ 36 124]]
```

18. WAP to demonstrate CNN for edge detection in Conv2D.

```
!pip install tensorflow
```

```
import tensorflow as tf
```

```
!pip install keras
```

```
from tensorflow import keras
```

```
!pip install matplotlib
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
!pip install opencv-python
```

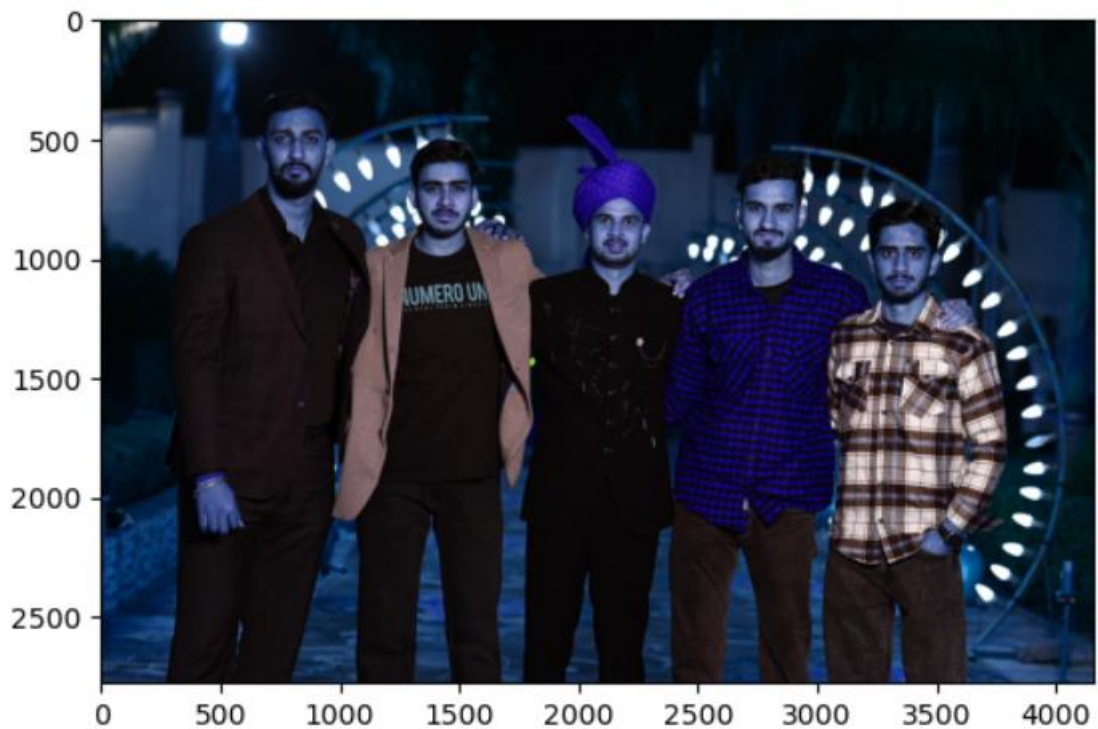
```
import cv2
```

```
img = cv2.imread("mlx.jpg")
```

```
print(img)
```

```
plt.imshow(img)
```

<matplotlib.image.AxesImage at 0x1ea008f4560>



```
from tensorflow.keras import layers
from tensorflow.keras.layers import Conv2D
newImg=Conv2D(filters=3,strides=(1,1),kernel_size=(3,3),padding="same"
,use_bias=False)
print(img.dtype)
print(img.shape)
img=tf.cast(img,tf.float32)/255
kernel=np.array([[0,-1,0],[-1,4,-1],[0,-1,0]],dtype=np.float32)
print(kernel.dtype)
img=np.expand_dims(l,axis=0)
print(img.shape)
print(kernel.shape)
kernel = kernel[:, :, np.newaxis, np.newaxis]
print(kernel.shape)
kernel = np.repeat(kernel, repeats=3, axis=2)
kernel=np.repeat(kernel, repeats=3, axis=3)
print(kernel.shape)
newImg.build(input_shape=img.shape)
newImg.set_weights([kernel])
```

```
output=newImg(img)[0]  
plt.imshow(output)
```

<matplotlib.image.AxesImage at 0x1ea009b6cc0>

